Lecture 4 -08/11

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Geometry 2:
Epipolar Geometry
two-view geometry

Introduction to Computer Vision

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The aim of geometry in computer vision

- Multi-view reconstruction
- Structure from Motion (SfM)

Given Images



4 images out of a sequence with 435 images.

Compute 3D Model



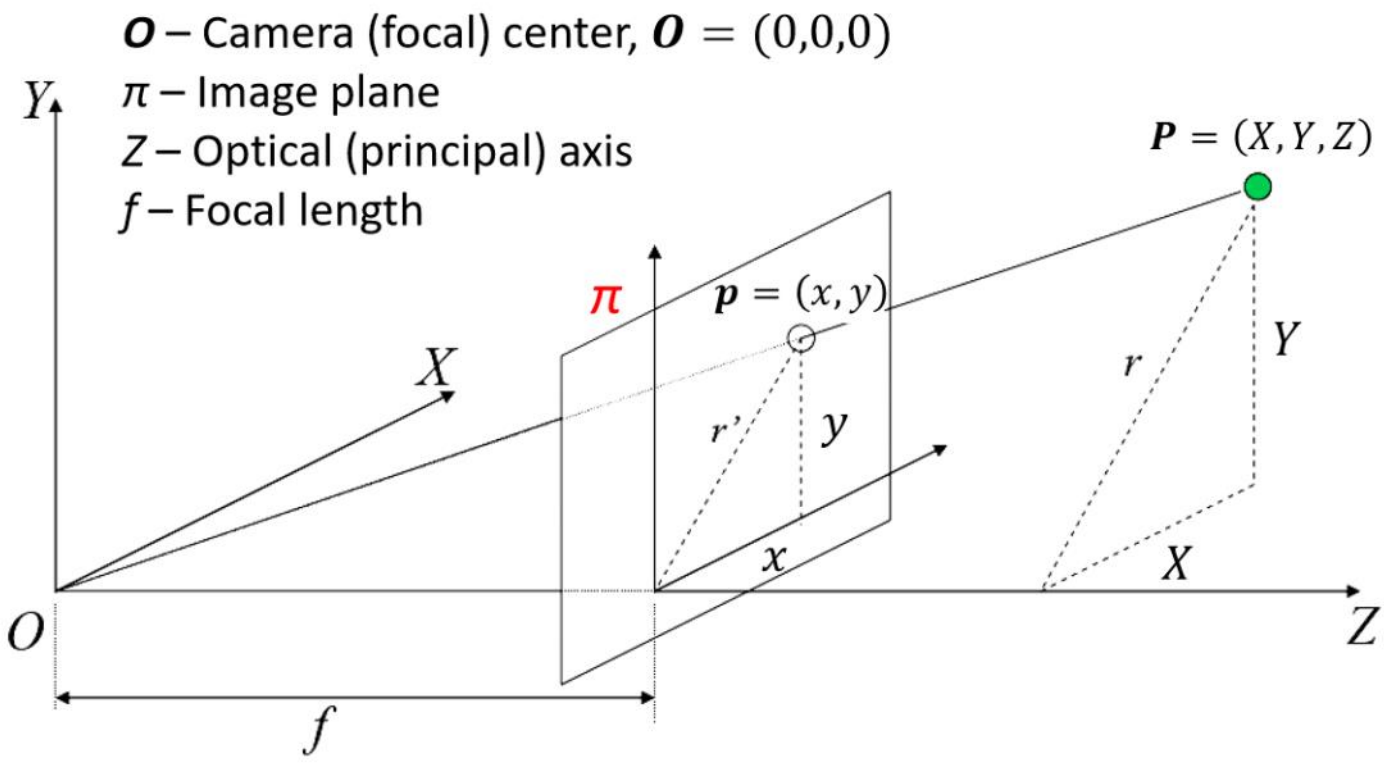
- Reconstruction of the 3D structure
- Camera pose recovery

Geometry - Material covered

- Pinhole camera model, perspective projection, projective geometry
- Two view geometry:
  - Epipolar geometry, the essential matrix
  - Camera calibration, the fundamental matrix
- Stereo vision
  - 3D reconstruction from two views
- Multi-view geometry
  - reconstruction through factorization

Summary

The pinhole camera - perspective projection

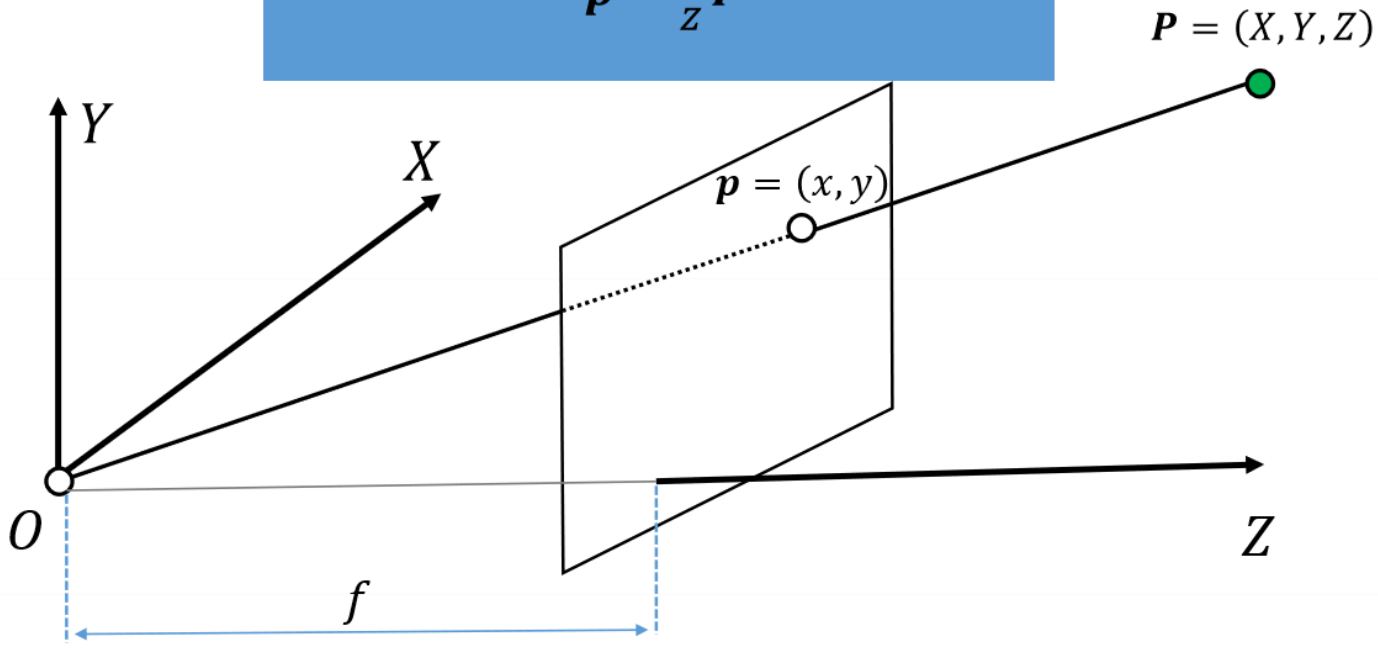


Summary

The pinhole camera - perspective projection

$$x = \frac{fX}{Z} \qquad y = \frac{fY}{Z} \qquad z = f$$

$$\boldsymbol{p} = \frac{f}{Z}\boldsymbol{P}$$



Summary

Perspective projection

- Perspective rule $P = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \rightarrow p = \begin{bmatrix} x \\ y \end{bmatrix}$

$$x = \frac{fX}{Z} \qquad y = \frac{fY}{Z}$$

- Using matrix notations and homogeneous coordinates

$$p = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \begin{bmatrix} f & & & 0 \\ & f & & 0 \\ & & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

- Camera matrix* $3D \rightarrow 2D$ $(P \rightarrow p)$

$$\begin{bmatrix} f & & & 0 \\ & f & & 0 \\ & & 1 & 0 \end{bmatrix}_{3\times 4} = \begin{bmatrix} f & & \\ & f & \\ & & 1 \end{bmatrix} [I|\mathbf{0}]$$

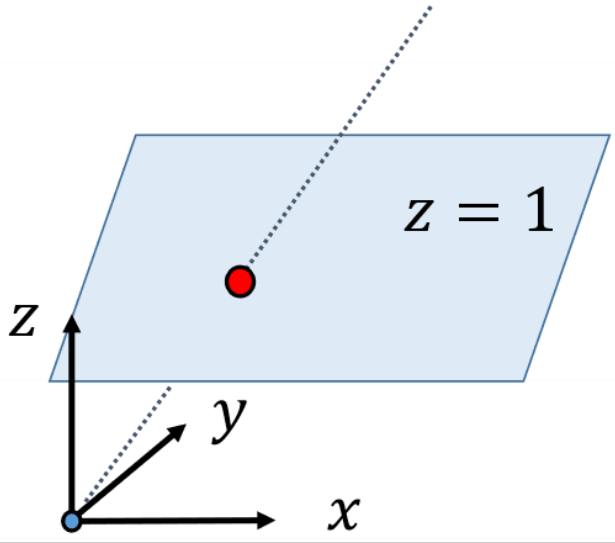
*Special case

Projective plane - points

- A point $\boldsymbol{p} \in \mathbb{P}^2$ is a (two sided) ray in $\mathbb{R}^3$

$$\boldsymbol{p} = \begin{pmatrix} \alpha x \\ \alpha y \\ \alpha \end{pmatrix} \quad \alpha \neq 0$$

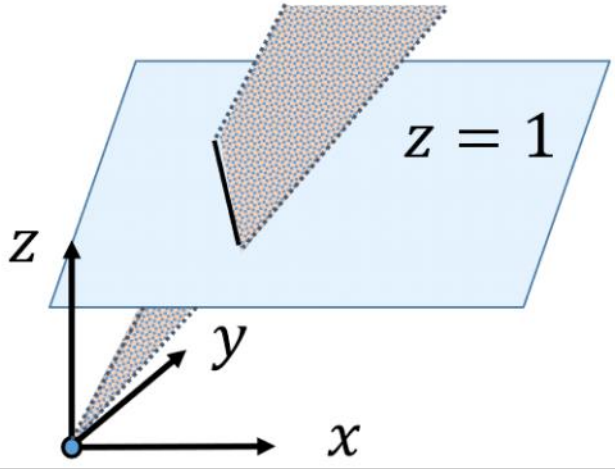
- $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$



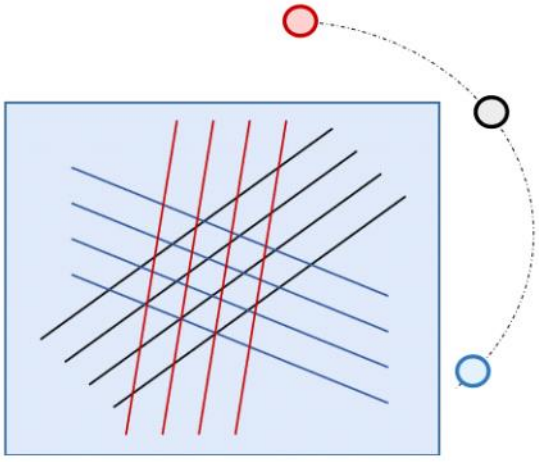
Projective geometry – lines

- A point $\boldsymbol{p} \in \mathbb{P}^2 \Leftrightarrow$ a (two sided) ray in $\mathbb{R}^3$

  $\alpha\boldsymbol{p}$ with $\alpha \neq 0$
- A line $\boldsymbol{l} \in \mathbb{P}^2 \Leftrightarrow$ a central plane in $\mathbb{R}^3$
- With representation $\boldsymbol{l} = k(a, b, c)^T, k \neq 0$
- A point $\boldsymbol{p} \in \mathbb{P}^2$ resides on a line $\boldsymbol{l} \in \mathbb{P}^2 \Leftrightarrow \boldsymbol{l}^T\boldsymbol{p} = 0$



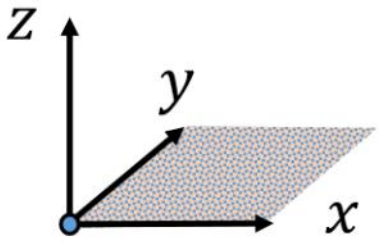
Projective geometry – vanishing (ideal) points



- Parallel lines intersect at ideal points

$$\boldsymbol{p} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

- Determined by the ratio $\frac{x}{y}$
- The complete set of ideal points is the line at infinity
- The line at infinity, $\boldsymbol{l}_\infty$, is the $Z = 0$ plane
- $\boldsymbol{l}_\infty = (0,0,1)^T$



Projective transformation = Projectivity = Homography

- A projectivity h is an invertible mapping from points in $\mathbb{P}^2$ to points in $\mathbb{P}^2$ that maps lines to lines

- $\boldsymbol{x_1}, \boldsymbol{x_2}$ and $\boldsymbol{x_3}$ lie on the same line$\Leftrightarrow h(\boldsymbol{x_1}), h(\boldsymbol{x_2})$ and $h(\boldsymbol{x_3})$ reside on the same line

- A homography h: $\mathbb{P}^2 \rightarrow \mathbb{P}^2$ is a non-singular matrix 3×3 matrix H

Homography transforms a line to a line

Projective transformation = Projectivity = Homography

- An invertible linear transformation of rays

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \boldsymbol{q} \propto H\boldsymbol{p} = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

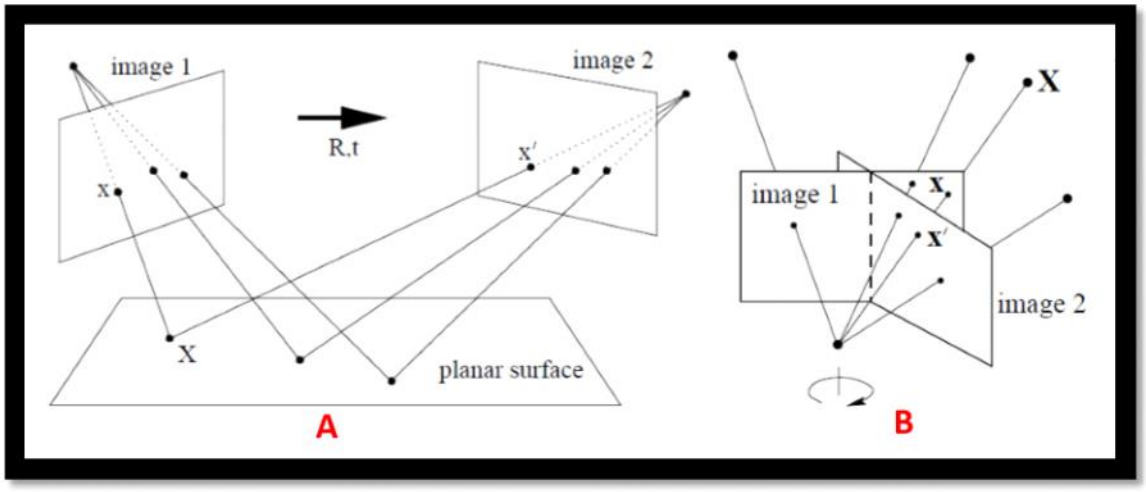
- Non-linear transformation of points

$$x' = \frac{[H\boldsymbol{p}]_1}{[H\boldsymbol{p}]_3} = \frac{h_{11}x+h_{12}y+h_{13}}{h_{31}x+h_{32}y+h_{33}}, \quad y' = \frac{[H\boldsymbol{p}]_2}{[H\boldsymbol{p}]_3} = \frac{h_{21}x+h_{22}y+h_{23}}{h_{31}x+h_{32}y+h_{33}}$$

- H is determined up to scale, 8 DOFs
- The unknowns are the 9 entries of the homography
- Homography can be calculated by Direct Linear Transform (DLT)
- How many points are needed to determine homography?
  - 4 point matches: each point match determines two equations

Summary

Examples of projective transformation



(A) The projective transformation between two images induced by a world plane

(B) The projective transformation between two images with the same camera center (camera rotating about its center, no translation)

- Re-proof in a few slides

Camera motion model external parameters

$$\begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = R \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \boldsymbol{t}$$

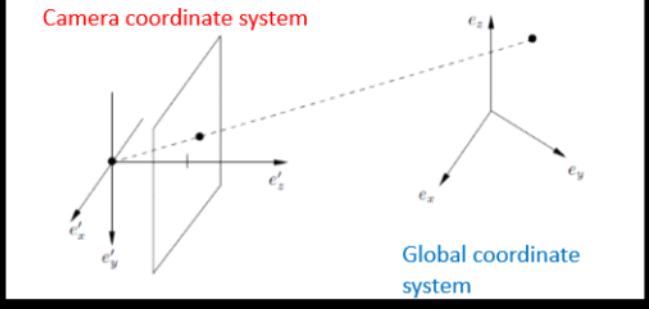
$R \in \mathbb{R}^{3\times3}$ rotation matrix

$R^T R = I, \ \det(R) = 1$

$\Rightarrow \mathrm{R}^{-1} = R^T$

$\boldsymbol{t} \in \mathbb{R}^3$ translation vector

Summary



Global coordinate system

$R = I, \boldsymbol{t} = \boldsymbol{0}$

$$X_3' \begin{bmatrix} X_1'/X_3' \\ X_2'/X_3' \\ 1 \end{bmatrix} = \begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = [R \ \boldsymbol{t}] \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ 1 \end{bmatrix}$$

← homogeneous coordinates of a 3D point

$[R \ t]_{3\times4}$ is called camera matrix

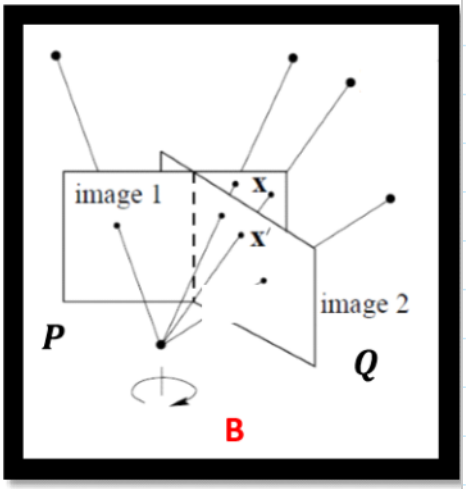
The rows of R encodes the x' axis, y' axis and z' axis of the local coordinate system, as expressed in the global coordinate system

Summary

(B) Camera rotation ➔ homography

$$\boldsymbol{Q} = R\boldsymbol{P} + \cancel{t}$$

Rotation around the focal center (no translation)



$\boldsymbol{p}, \boldsymbol{q}$ are perspective projections of $\boldsymbol{P}, \boldsymbol{Q}$

Aim: Find invertible H s.t. $\boldsymbol{q} \propto H\boldsymbol{p}$

$$\boldsymbol{p} = \frac{f}{Z}\boldsymbol{P} \Rightarrow \boldsymbol{P} = \frac{z}{f}\boldsymbol{p}$$

$$\boldsymbol{q} = \frac{f}{Z'}\boldsymbol{Q} \Rightarrow \boldsymbol{Q} = \frac{z'}{f}\boldsymbol{q}$$

Substitute in $\boldsymbol{Q} = R\boldsymbol{P}$

$$\frac{z'}{f}\boldsymbol{q} = \frac{z}{f}R\boldsymbol{p}$$

$$\Rightarrow \boldsymbol{q} = \frac{Z}{Z'}R\boldsymbol{p}$$

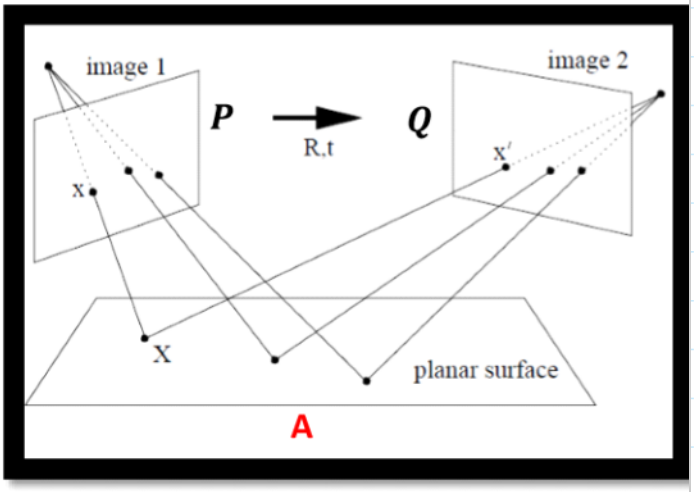
$$\boldsymbol{P} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad \boldsymbol{Q} = \begin{bmatrix} X' \\ Y' \\ Z' \end{bmatrix}$$

$$\Rightarrow \boldsymbol{q} \propto R\boldsymbol{p}, \text{ with } H = R$$

(⇒The 3D information, i.e. the depth, cannot be recovered)

(A) Planar scene ➔homography

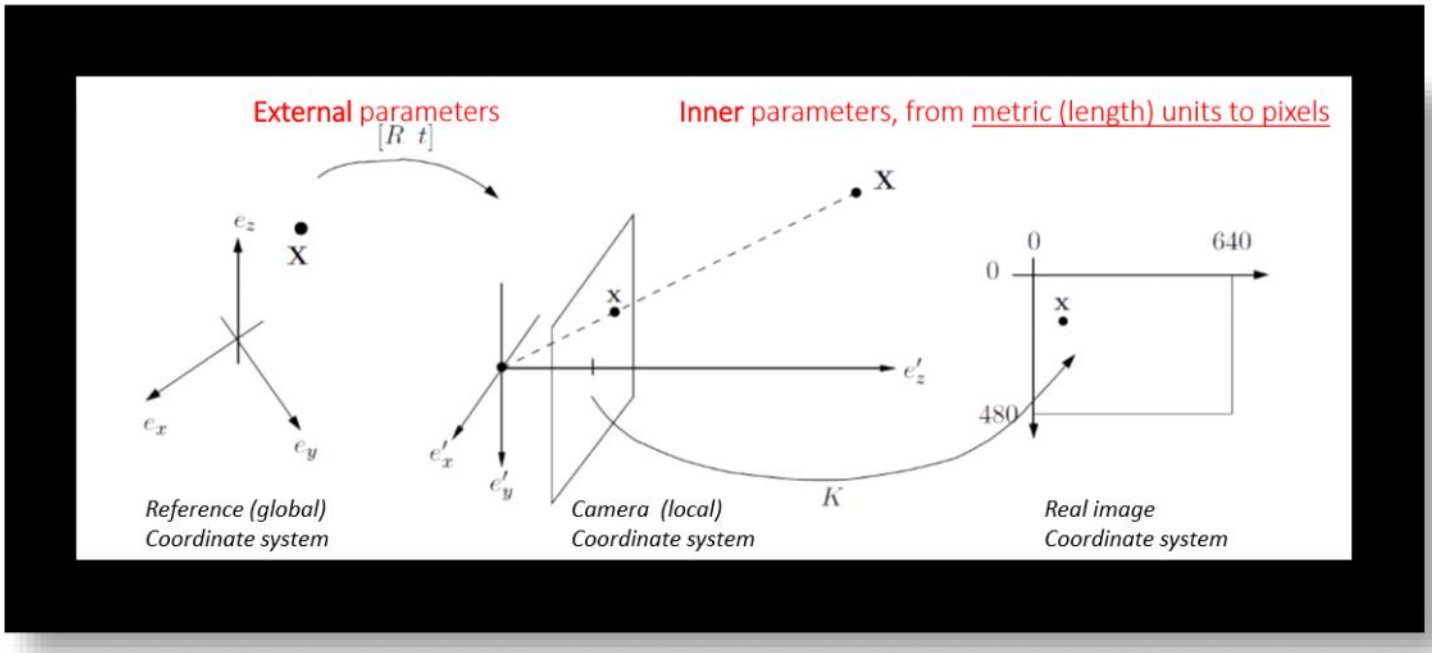
- A plane $aX + bY + cZ = d$
- In vector notation $\boldsymbol{n}^T \boldsymbol{P} = d$
- $\boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}$
- $\frac{\boldsymbol{n}^T \boldsymbol{P}}{d} = 1$, so $\boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}\frac{\boldsymbol{n}^T \boldsymbol{P}}{d}$
- Rearrange, $\boldsymbol{Q} = \left(R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T\right)\boldsymbol{P}$
- Project, $\boldsymbol{q} \propto \left(R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T\right)\boldsymbol{p}$
- So $\boldsymbol{q} \propto H\boldsymbol{p}$, with $H = R + \frac{1}{d}\boldsymbol{t}\boldsymbol{n}^T$



Hierarchy of transformations

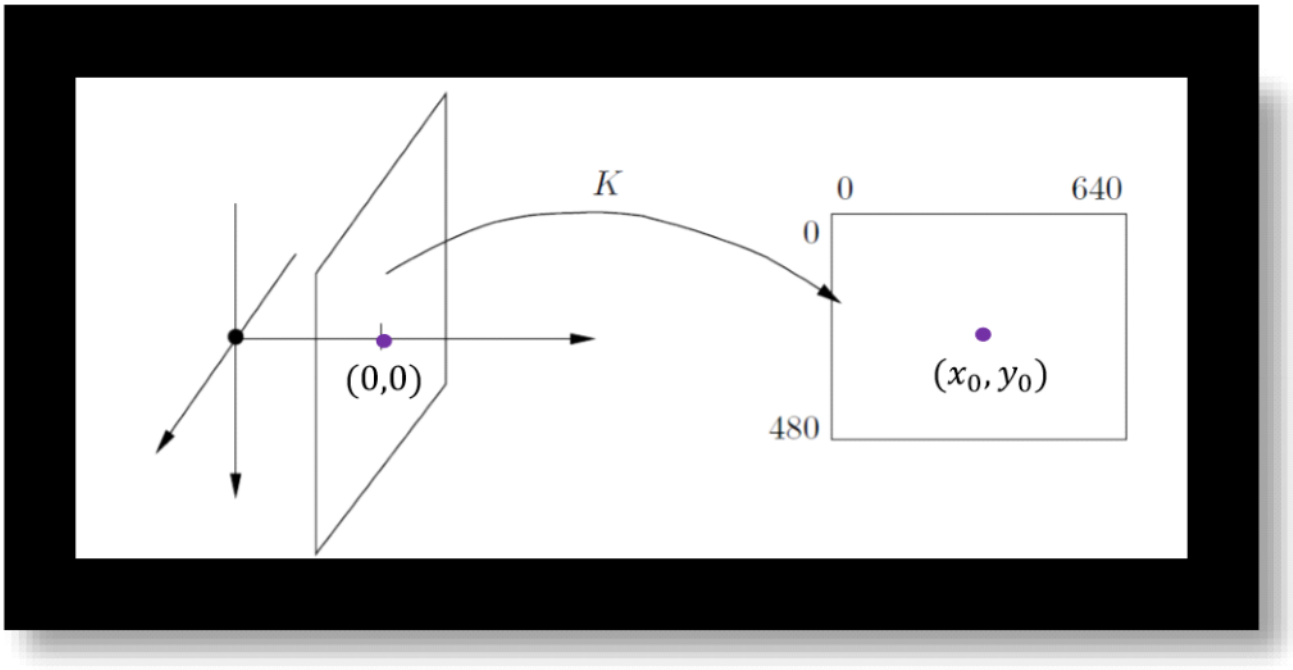
Rigid	$\begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, lengths, area, parallelism
Similarity	$\begin{pmatrix} a & -b & t_x \\ b & a & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves angles, parallelism
Affine	$\begin{pmatrix} a & b & t_x \\ c & d & t_y \\ 0 & 0 & 1 \end{pmatrix}$	Preserves parallelism
Homography	$\begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix}$	Preserves cross ratio

The role of the inner parameters K of a camera

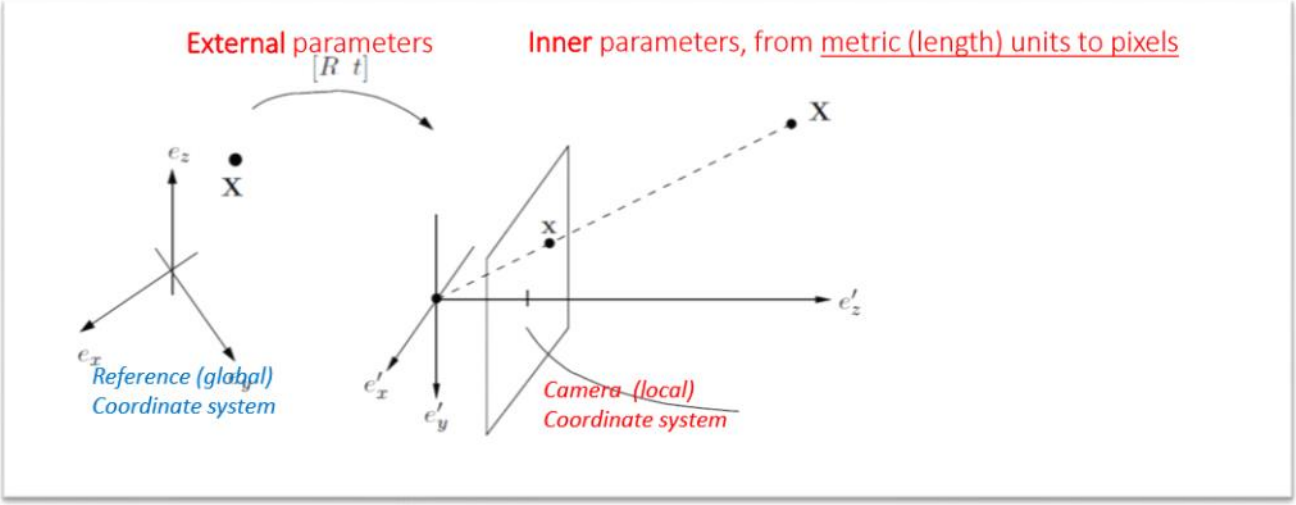


The (linear) mapping K from the image plane to the real image

The role of the inner parameters K of a camera



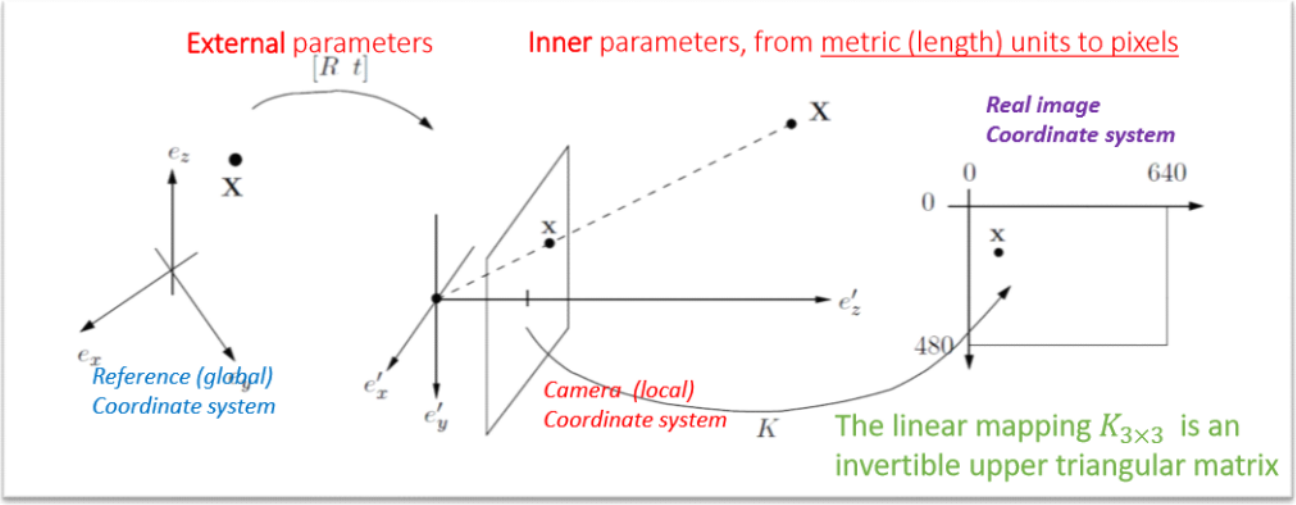
The (linear) mapping K from the image plane to the real image



- **Calibrated case, metric space**

$$\lambda \boldsymbol{x} = \underbrace{[R\ \boldsymbol{t}]}_{C} \underbrace{\begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ 1 \end{pmatrix}}_{\boldsymbol{X}}$$

- $C = [R\ \boldsymbol{t}]_{3\times4}$ calibrated camera matrix
- $\boldsymbol{x} = \begin{bmatrix} \boldsymbol{x_1} \\ \boldsymbol{x_2} \\ \boldsymbol{1} \end{bmatrix}$ calibrated coordinates



- **Uncalibrated case**

$$\lambda \boldsymbol{x} = \underbrace{K[R\ \boldsymbol{t}]}\begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ 1 \end{pmatrix}$$

- $C = K[R\ \boldsymbol{t}]$ uncalibrated camera matrix, also of size 3×4
- $\boldsymbol{x} = \begin{bmatrix} \boldsymbol{x_1} \\ \boldsymbol{x_2} \\ \boldsymbol{1} \end{bmatrix}$ uncalibrated coordinates

Camera matrix $C_{3\times4} = K[R|t]$

- Camera matrix $C_{3\times4}$ captures camera translation $\boldsymbol{t}$, orientation R, and (linear) internal calibration parameters K

$$\boldsymbol{x} \propto K[R|t]\boldsymbol{X}$$

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \propto \underbrace{\begin{pmatrix} \alpha_x & \gamma & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Internal calibration}} \underbrace{\begin{pmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{pmatrix}}_{\text{external calibration}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

The linear mapping $K_{3\times3}$ is an invertible upper triangular matrix

- As before, '$\propto$' means "up to (non-zero) scale factor"
  Scale is different for every point

Calibration matrix K

- A 3×3 upper triangular matrix, K, that captures (linear) internal calibration parameters

$$K = \begin{pmatrix} \alpha_x & \gamma & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Parameters:
  - (x_0, y_0) - image center (in pixel dimensions)
  - $(\alpha_x, \alpha_y) = (fm_x, fm_y)$
  - f - focal length
  - (m_x, m_y) – number of pixels per unit distance in x and y directions
  - γ – skew (for tilted pixels, typically 0)
- Radial distortions are treated separately
- Both linear and radial calibration parameters are available in Exif tags

The role of the calibration matrix K of a camera C

$$K = \begin{pmatrix} \alpha_x & \gamma & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$$

- K transforms from the image plane (calibrated) to the real image (pixels)
- K^{-1} maps from the real image (pixels) to the image plane
- Calibration parameters are available in EXIF tags

Camera matrix

$$C = K[R|\boldsymbol{t}]$$

- The camera matrix is of size 3×4
- How many degrees of freedom (dof)?
- $R \in \mathbb{R}^{3\times3}$ rotation matrix, 3 dof
- $\boldsymbol{t} \in \mathbb{R}^3$ translation vector, 3 dof
- $K = \begin{pmatrix} \alpha_x & \gamma & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$ calibration matrix, 5 dof

$\Rightarrow$ 11 dof (out of 12 entries)

Any multiplication by a non-zero scalar results in an equivalent camera matrix

Camera matrix

Calibrated camera

- Camera equations

$$\lambda \boldsymbol{x} = C\boldsymbol{X}$$

- Def. A camera $C = K[R\ \boldsymbol{t}]$ is called calibrated if the inner parameters K are known.
- We can multiply the camera equations from both sides by K^{-1}, yielding

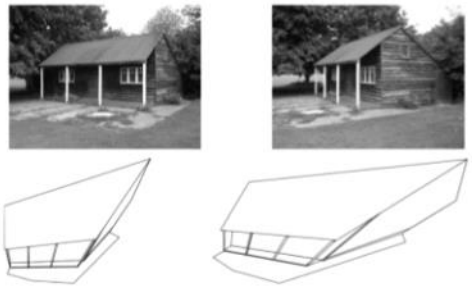
$$\lambda K^{-1}\boldsymbol{x} = [R\ \boldsymbol{t}]\boldsymbol{X}$$
$$\lambda \tilde{\boldsymbol{x}} = [R\ \boldsymbol{t}]\boldsymbol{X}$$

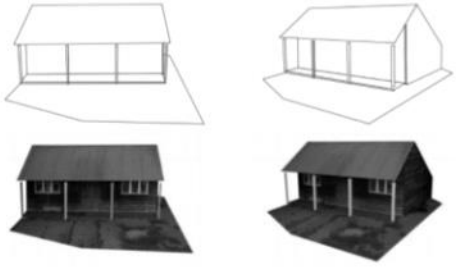
with $\tilde{\boldsymbol{x}} = K^{-1}\boldsymbol{x}$

- Def. The camera matrix $[R\ \boldsymbol{t}]$ is called normalized (calibrated) camera and $\tilde{\boldsymbol{x}}$ are called the normalized (calibrated) image points

Projective reconstruction vs. Euclidean reconstruction

Uncalibrated reconstruction vs. Calibrated reconstruction





Projective Reconstruction

The uncalibrated case

Do not preserve angles and parallel lines

Euclidean reconstruction

The calibrated case

Preserve angles and parallel lines

Projective reconstruction vs. Euclidean reconstruction

Uncalibrated reconstruction vs. Calibrated reconstruction

- Trying to build a 3D model from internet or photos taken by unknown cameras
- We do not know ahead of time the intrinsic calibration
- In such cases the reconstruction may look weird
- Orthogonal lines or planes in the world may not end up being reconstructed as orthogonal
- The distinction between calibrated camera and uncalibrated camera is tightly connected to the degrees of freedom of the reconstruction space
- Calibrated- up to similarity transformation
- Uncalibrated- up to projective transformation
- Nevertheless, in such situations we can still have the Epipolar Geometry between two views

Epipolar geometry

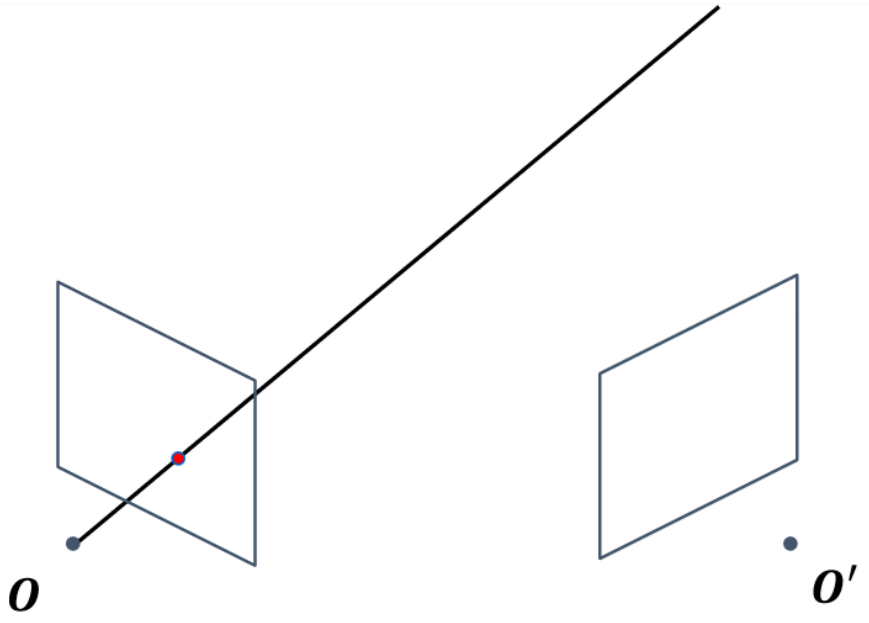
The *epipolar geometry* is the geometry between two views, also called stereo vision

When two cameras view a 3D scene from two distinct positions, there are a number of geometric relations between the projection of the 3D points onto the 2D images that lead to constraints between the image points



Epipolar geometry

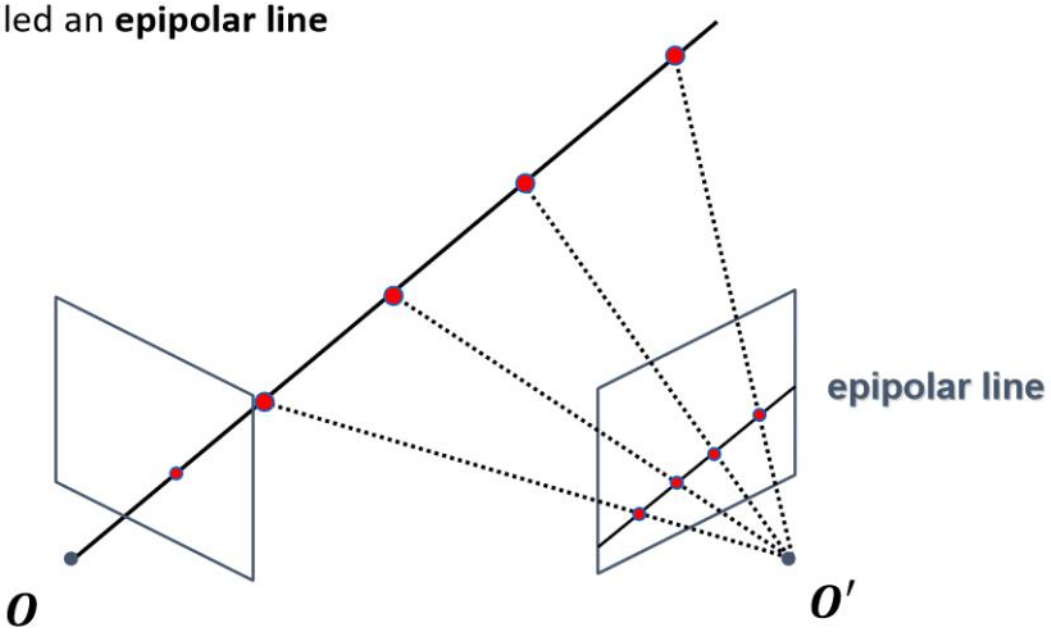
Let's look at a point at the left image
Without knowing its depth, it can come from any 3D on its back-projected ray



Epipolar geometry

A corresponding point in the right image should lie on the projection of this 3D ray to the right image

This is called an **epipolar line**

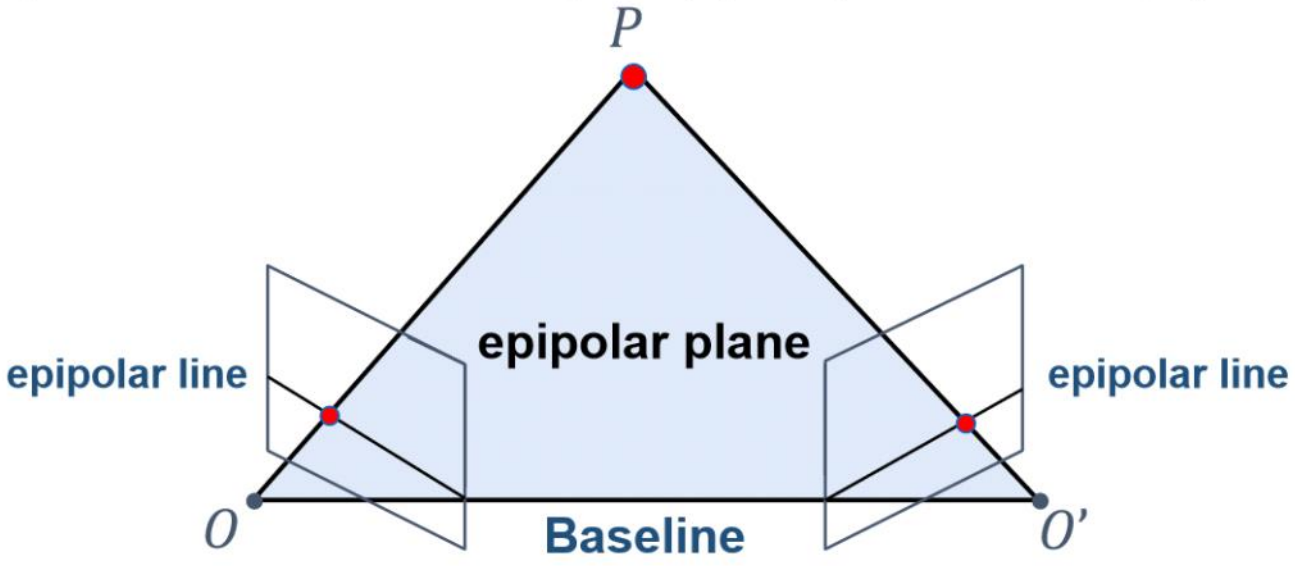


Epipolar geometry

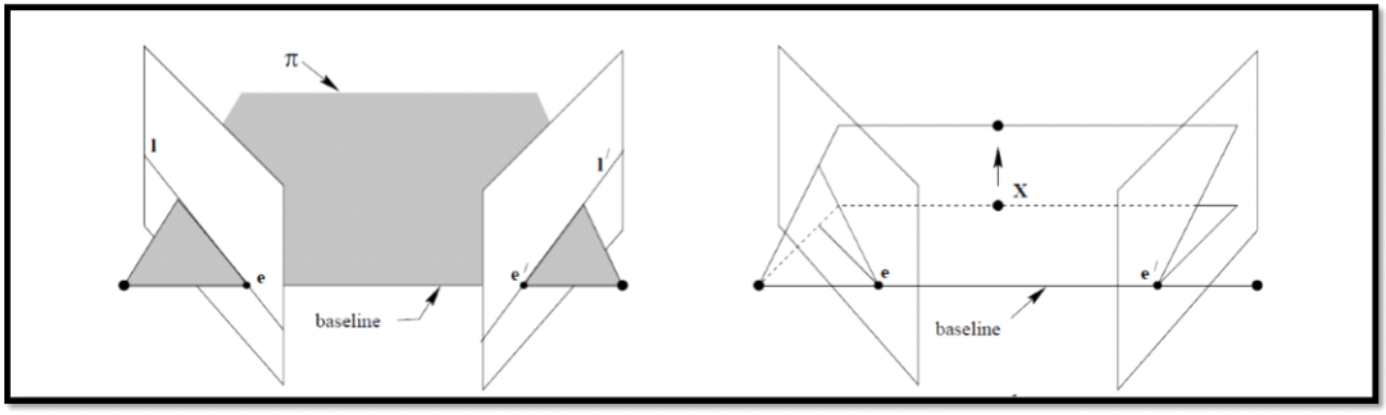
Baseline: the line joining the camera centers

Epipolar plane: a plane that contains the baseline

Epipolar line: *an intersection of an epipolar plane with image plane*



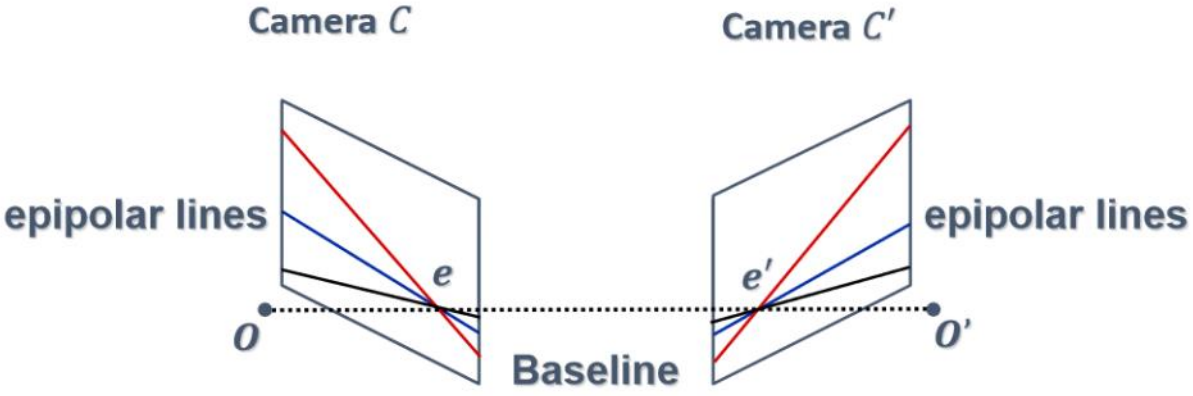
Epipolar geometry



- *The **epiplar planes** rotate about the baseline*
- *Any **epipolar plane π** intersects the image plane at **epipolar lines l, l′***
- *The camera center resides on any ray which is projected to the other view. Therefore, the projection of the camera center (**epipole**) should reside on each **epipolar line***
- *The **camera baseline** intersects each image plane at the **epipoles e, e′***
- ***All epipolar lines intersect at the epipole***

Epipolar geometry

- An **epipole** is the projection in one view of the camera center of the other view
- The epipole $\boldsymbol{e}$ is the projection of $\boldsymbol{O'}$ to the left image (using C)
- The epipole $\boldsymbol{e'}$ is the projection of $\boldsymbol{O}$ to the right image (using C')
- $\boldsymbol{e}, \boldsymbol{e'}$ are usually outside the field of view



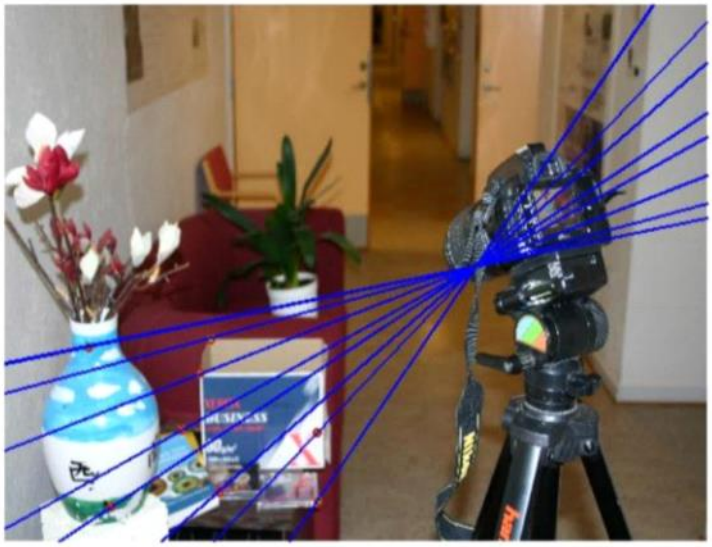
Epipolar geometry – example





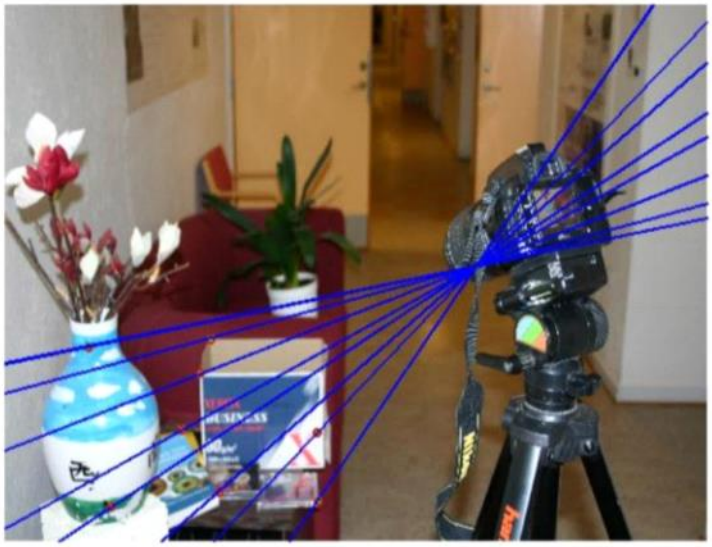
Epipolar geometry – example





Epipolar geometry – example





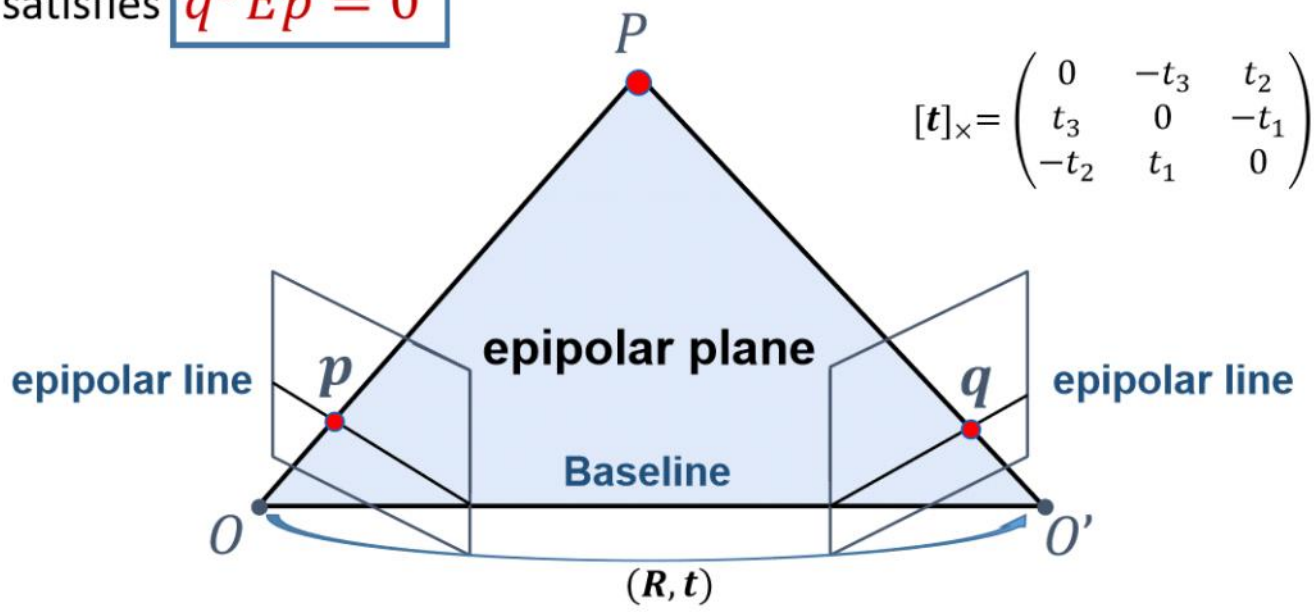
The projected lines should all meet in a point. The so called epipole is the projection of the camera center of the other camera.

Epipolar geometry – the Essential matrix

The calibrated case

- $\boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}, \quad \boldsymbol{p} = \frac{f}{Z}\boldsymbol{P}, \quad \boldsymbol{q} = \frac{f'}{Z'}\boldsymbol{Q}$
- The Essential matrix $E = [\boldsymbol{t}]_{\times} R$
- satisfies $q^T E p = 0$

$$[\boldsymbol{t}]_{\times} = \begin{pmatrix} 0 & -t_3 & t_2 \\ t_3 & 0 & -t_1 \\ -t_2 & t_1 & 0 \end{pmatrix}$$



Epipolar geometry – the Essential matrix

The calibrated case

$$\boldsymbol{p} = \frac{f}{Z}\boldsymbol{P}, \quad \boldsymbol{q} = \frac{f'}{Z'}\boldsymbol{Q}, \quad \boldsymbol{Q} = R\boldsymbol{P} + \boldsymbol{t}$$

$$\boldsymbol{t} \times \boldsymbol{Q} = \boldsymbol{t} \times R\boldsymbol{P} \quad (\boldsymbol{t} \times \boldsymbol{t} = 0)$$

$$0 = \boldsymbol{Q}^T(\boldsymbol{t} \times \boldsymbol{Q}) = \boldsymbol{Q}^T(\boldsymbol{t} \times R\boldsymbol{P}) = \boldsymbol{Q}^T[\boldsymbol{t}]_\times R\boldsymbol{P}$$

$$0 = \frac{z'z}{f'f}\boldsymbol{q}^T[\boldsymbol{t}]_\times R\boldsymbol{p} = \boldsymbol{q}^T\underbrace{[\boldsymbol{t}]_\times R}\boldsymbol{p}$$

$$\boldsymbol{q}^T E\boldsymbol{p} = 0$$

$E = [\boldsymbol{t}]_\times R$ is called the *Essential* matrix

The Essential matrix, $\boldsymbol{q}^T E\boldsymbol{p} = 0$

$$E = [\boldsymbol{t}]_\times R$$

- Algebraically, homogeneous quantity⇒scale ambiguity

The calibrated case

- Rank(E)=2, right and left null spaces of the E = epipoles

- Two singular values are equal and the third is 0

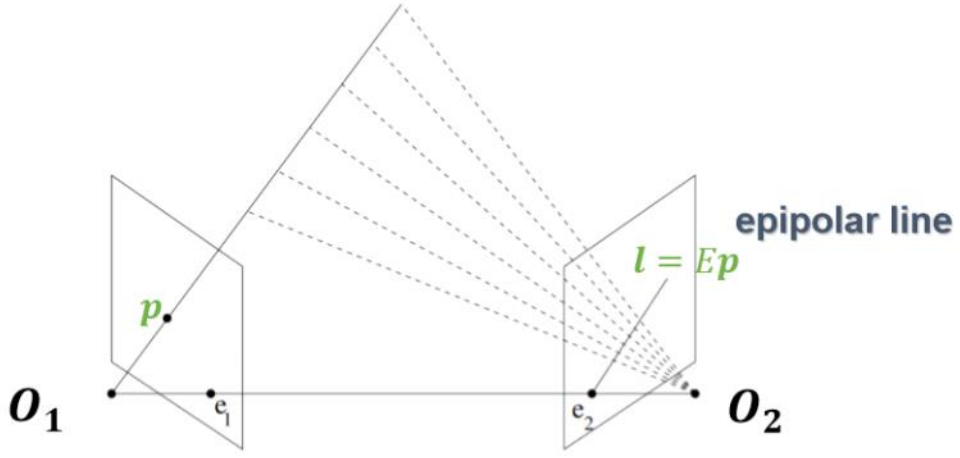
- $E = [\boldsymbol{t}]_\times R$, 5 DOFs:
  - Translation (up to scale) (2 dofs)
  - Rotation (3 dofs)

The Essential matrix, $\boldsymbol{q^T E p = 0}$

$$E = [t]_\times R$$

The calibrated case

- Geometrically…
- The essential matrix $E \in \mathbb{R}^{3\times3}$ transforms each point $\boldsymbol{p} \in \mathbb{P}^2$ to its corresponding epipolar line $\boldsymbol{l = Ep} \in \mathbb{P}^2$ in the other image



The Essential matrix, $\boldsymbol{q}^T E\boldsymbol{p} = 0$

The calibrated case

Estimating E

- point matches between the two images are needed
  - Linear solution (DLT) requires (at least) 8 matches
  - Non-linear (polynomial) solution requires 5 matches

Recovering t and R from E (camera pose)

- 4 solutions (two orthogonal matrices - rotation and reflection, sign ambiguity for translation).
  Resolve ambiguity: positive depths in both images

Recovering camera pose is an important step towards structure from motion problem

Epipolar geometry - the Fundamental matrix

The uncalibrated case

- The mapping between the image plane (Euclidean space) and the real image (pixels), is given by…
- Calibration matrix K
- The relation between the calibrated $(\tilde{p}, \tilde{q})$ and the uncalibrated coordinates (p, q) is given by

$$p = K\tilde{p},\ q = K\tilde{q}$$

$$\Rightarrow 0 = \tilde{q}^T E\tilde{p} = q^T K^{-T} E K^{-1} p$$

- Let

$$F = K^{-T} E K^{-1}$$

- Then

$$q^T F p = 0$$

- F is called the Fundamental matrix

The Fundamental matrix, $q^T Fp = 0$

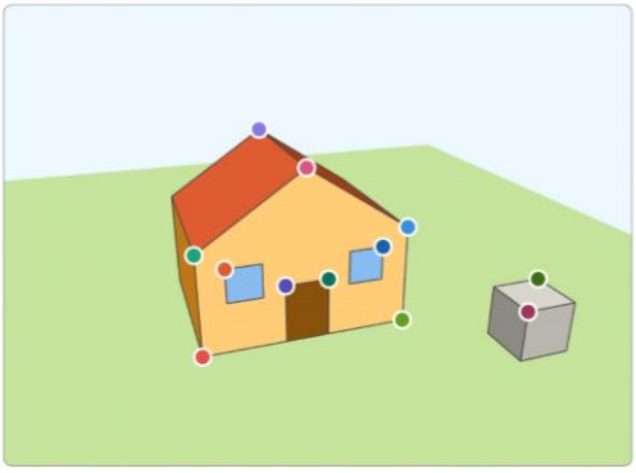
The uncalibrated case

- Similar to the mapping obtained by the Essential matrix, the Fundamental matrix F maps from a point in one image to a line in the second image

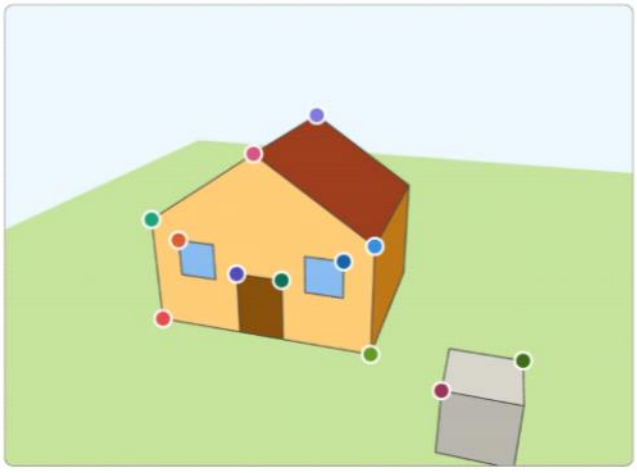
- $F \in \mathbb{R}^{3\times3}$ has 7 dof
  - F is defined up to scale
  - Rank (F)=2 (null spaces contain epipoles)

- Recovering F:
  - Linear solution requires 8 pairs of corresponding points
  - Non-linear solution requires 7 pairs: rank 2 is enforced by requiring $\mathrm{Det}(F) = 0$

Estimating the Fundamental (Essential) matrix F

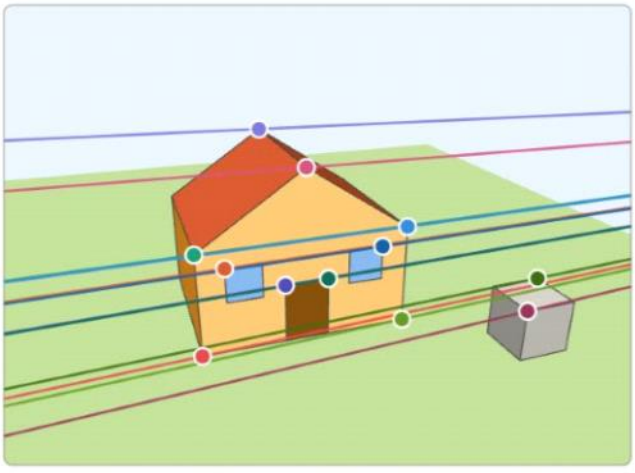


View 1 (features p)

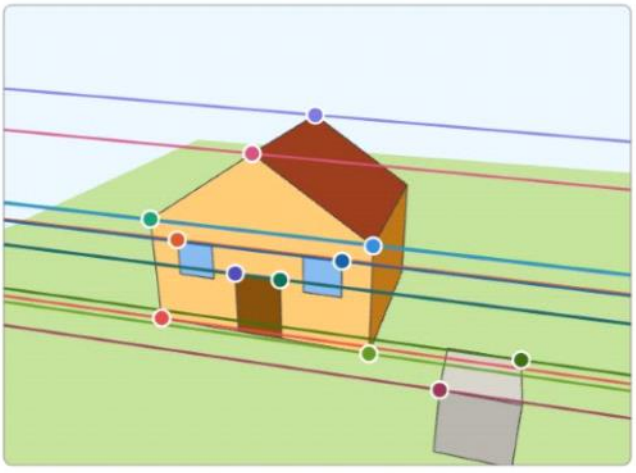


View 2 (features q)

Estimating the Fundamental (Essential) matrix F



View 1: points p, lines $l = F^T q$



View 2: points q, lines $l' = Fp$

Estimating the Fundamental (Essential) matrix F

We have that if $\boldsymbol{p} \leftrightarrow \boldsymbol{q}$ are corresponding points

$$\Rightarrow \boldsymbol{q}^T F \boldsymbol{p} = 0$$

If $\boldsymbol{p} = (p_1, p_2, p_3)^T$, $\boldsymbol{q} = (q_1, q_2, q_3)^T$

$$\boldsymbol{q}^T F \boldsymbol{p} = (q_1, q_2, q_3) \begin{pmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} =$$

$$F_{11}q_1p_1 + F_{12}q_1p_2 + F_{13}q_1p_3 + F_{21}q_2p_1 + F_{22}q_2p_2 + F_{23}q_2p_3 + F_{31}q_3p_1 + F_{32}q_3p_2 + F_{33}q_3p_3 = 0$$

A linear equation in F's elements

Estimating the Fundamental (Essential) matrix by Direct Linear Transform (DLT) algorithm

Linear solution of $\{\boldsymbol{q}^T F \boldsymbol{p} = 0\}$

$$\begin{pmatrix} q_1^1 p_1^1 & q_1^1 p_2^1 & \cdots & q_3^1 p_3^1 \\ q_1^2 p_1^2 & q_1^2 p_2^2 & & q_3^2 p_3^2 \\ \vdots & & & \vdots \\ q_1^8 p_1^8 & q_1^8 p_2^8 & \cdots & q_3^8 p_3^8 \end{pmatrix} \begin{pmatrix} F_{11} \\ \\ \vdots \\ F_{33} \end{pmatrix} = 0$$

- Find null space with SVD (Singular Value Decomposition)
- *Data normalization is an essential step in the DLT algorithm, it must not be considered optional*
- $n > 8$ point correspondences we use homogeneous least squares solution

Direct Linear Transform (DLT) algorithm

Matrix formulation

$$M\boldsymbol{v} = 0$$

We are interested in $0 \neq \boldsymbol{v} \in null(M)$ (arbitrary scale)

In general, due to noise in the measurements, we will not have an exact solution

We look for a solution for the optimization problem

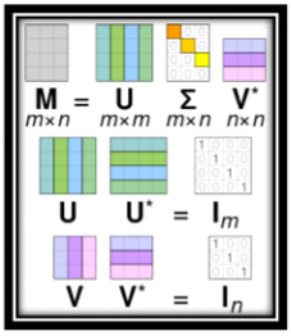
Homogeneous least-squares problem

$$\min_{\|\boldsymbol{v}\|=1} \|M\boldsymbol{v}\|$$

Solution

$\boldsymbol{v}$ is the last column of V, where $M = UDV^T$ is the SVD of M

Singular Value Decomposition (SVD)



- Theorem. A matrix $M \in \mathbb{R}^{m \times n}$ can be factorized into

$$M = UDV^T$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices and $D = diag(\sigma_1, \dots, \sigma_p) \in \mathbb{R}^{m \times n}$ with $p = \min\{m, n\}$ where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$, called singular values.

- It can be shown that if we define r by

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = \sigma_p = 0$$

then

$$\text{rank}(M) = r$$
$$\text{null}(M) = \text{span}\{v_{r+1}, \dots, v_n\}$$
$$\text{Img}(M) = \text{span}\{u_1, \dots u_r\}$$

Estimating the Fundamental matrix F

Issue:

- Resulting F may not have $\det(F) = 0$

Solution:

- Pick the closest matrix F with $\det(F) = 0$
- This can be done simply using SVD
  - $[U, S, V] = \text{svd}(F)$
  - $S(3,3) = 0$
  - $F = U * S * V^T$

Estimating the Fundamental matrix F

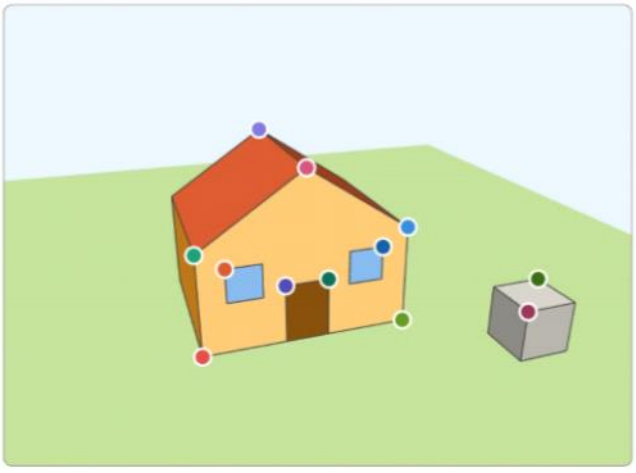
The 8-point algorithm:

- Extract at least 8 point correspondences
- Normalize the coordinates
- Form A and solve using SVD:

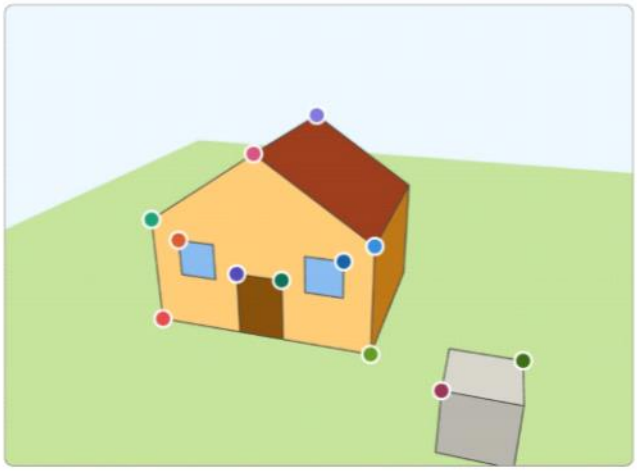
$$\min_{\|v\|=1} \|A\boldsymbol{v}\|^2$$

- Form the matrix F (ensure that $\det(F) = 0$)
- Transform back to the original coordinates

Estimating the Fundamental (Essential) matrix F

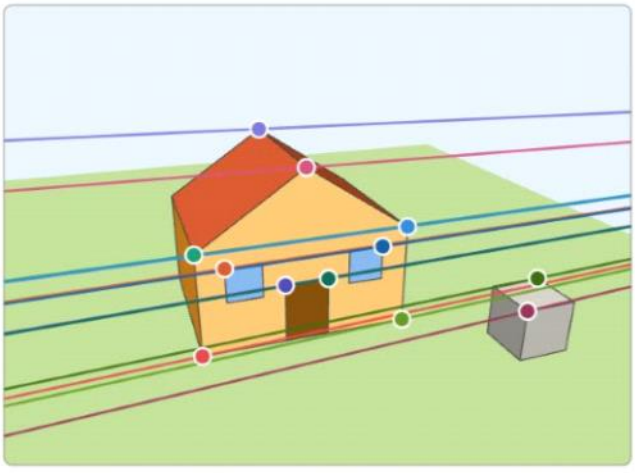


View 1 (features p)

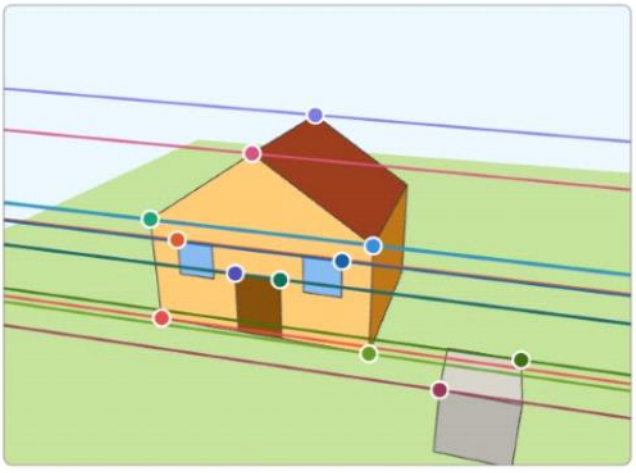


View 2 (features q)

Estimating the Fundamental (Essential) matrix F



View 1: points p, lines l = $F^T q$



View 2: points q, lines l' = Fp